

Cryptocurrencies: Who is vulnerable and what are the vulnerabilities?

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Abstract

Australians are embracing new forms of digital finance products and services, which includes purchasing cryptocurrencies and non-fungible tokens (NFTs). There has been an increase in investment scams associated with cryptocurrencies. In this article, we sought to understand from cryptocurrency and NFT investors, who is vulnerable and what vulnerabilities exist. We surveyed 745 Australians aged 18 and over who have purchased cryptocurrencies or NFTs. We used sociological perspectives of consumer vulnerability that focus on internal and external factors to analyse our findings. We found that both socioeconomic advantaged and disadvantaged Australians are vulnerable. The vulnerabilities include concerns over security, unsolicited advice, limited options for learning, and insufficient financial and IT literacy. The findings suggest that online financial education is needed from trusted independent sources to help combat scams and to keep Australians and their crypto assets safe. We recommend that more opportunities to educate individuals about alternative forms of financial products are offered in compulsory, vocational and higher education settings.

KEYWORDS

digital technology, education, financial regulation, financial services, investment

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1 | INTRODUCTION

In Australia, over \$2.7 billion was lost to scams in 2023 (Australian Government, 2024). Among the losses, investment scams topped the list, with \$1.3B (AUD) lost, and of that, \$171.1M in reported losses related to cryptocurrency investment scams (Australian Government, 2024). The increasing number of cryptocurrency related investment scams is not surprising as many Australians have reported that they do not know how to purchase cryptocurrencies (Morris, 2021) nor non-fungible tokens (NFTs) and likely do not know how to safely store them.

Cryptocurrencies are defined as “a new form of digital money” but “they aren't attached to a central bank, a country, or a regulatory body” (Danial et al., 2022, p. 7 &15) and are purchased from cryptocurrency exchanges. A NFT is defined as a “unique digital identifier that's secured and stored on a public blockchain. One token is not interchangeable for another, and a token cannot be further divided” (Laurence & Kim, 2021, p. 9). Not understanding a financial product and/or service can increase an individual's risk of being exploited or falling victim to a scam or fraud due to their lack of knowledge (O'Connor et al., 2019).

Education is seen as a preventative measure to reduce the risk of falling victim to scams. With over one million Australians owning cryptocurrencies (also referred to as crypto assets) (Morgan, 2022) and that number growing, the need to educate remains essential. However, Blue and Pinto (2021) report that not succumbing to financial scams, traps and/or exploitation is a challenge even the most financially literate individuals face.

The rapidly changing landscape of personal finance, in concert with trust issues in the finance sector, has created a perfect storm for vulnerable Australians navigating personal finances online. Contributing factors include the following: (1) the financial crisis of 2008, which saw a rise in the use of digital money and the emergence of Bitcoin (Ferguson, 2019); (2) a royal commission into the financial services sector in 2019; and (3) the COVID-19 global pandemic (2020 and ongoing) that pushed Australians to abandon cash (Caswell et al., 2020). Australians are now faced with a plethora of new online digital financial products and services, available via finance apps, to tempt, lure and/or exploit them, including cryptocurrencies and NFTs, which are the focus of this article.

In this paper, we are interested in individuals who are vulnerable in financial contexts due to a wide range of factors. By vulnerable, we are not just referring to individuals who may be living on low incomes and/or have insecure employment, despite it being widely recognised that individuals on low incomes are more vulnerable (Salignac et al., 2019). Instead, we are including individuals who may be financially vulnerable due to their lack of knowledge (Salignac et al., 2020) and due to their economic status (Hallegatte, 2014); income and employment status (Donnellan et al., 2009); access to resources, racial identity, age (Blue, Xing, & Pham, 2024); and education levels and disability (Morrow, 2008). We also agree with Menzel Baker et al. (2005, p. 137) who state that “consumer vulnerability is a condition, not a status”. They also state that some individuals may be more likely to experience vulnerability but that does not mean that every individual who holds the same demographic characteristics is vulnerable.

2 | INTERNAL AND EXTERNAL FACTORS OF VULNERABILITY IN A FINANCIAL CONTEXT?

Vulnerability is said to have both internal and external factors (Menzel Baker et al., 2005). Internal factors may be the demographics of the individual (Smith & Copper-Martin, 1997) such as language ability (Penãloza, 1995) and external factors may be the lack of access to goods and services (Hill, 2001). External factors are known to “contribute to imbalances of power in exchange relationships that disfavor consumers” (Menzel Baker et al., 2005, p. 130).

Understanding the role both internal and external factors play in vulnerability is a focus of this paper. We focus on these internal and external vulnerabilities that apply to owning cryptocurrencies and/or NFTs such as the demographics of crypto owners, their financial literacy/crypto literacy, their understanding of financial tax consequences and their ability to store crypto and/or NFTs. Next, we share what is known about cryptocurrencies owners.

2.1 | Internal factor—Demographics of cryptocurrencies owners

Information about cryptocurrency owners is reported to be scarce; however, Steinmetz et al. (2021) state that most owners are male and young. Weiom (2019) determined that Bitcoin owners, in addition to being male, are also university-educated, live in major cities and are active on the internet. Steinmetz et al. (2021) also reported that they are on higher incomes than non-owners. Other studies have shown no significant differences in the age of cryptocurrency users (Exton & Doidge, 2018), whereas Laboure and Reid (2020) report that those aged 18–34 have bought and sold cryptocurrencies at the highest levels. Last, Delfabbro et al. (2021) report that cryptocurrency trading is the fastest growing market in the world, which often attracts young, educated males on high incomes.

2.2 | Internal factor—The financial literacy of cryptocurrency owners

Lusardi (2019) argues that today more than ever, being financially literate is just as important as the ability to read and write. One reason being financially literate is so essential is that individuals are responsible now more than ever to look after their finances to ensure they have enough funds for a comfortable retirement (Lusardi, 2019). They also found that, an individuals' financial literacy level impacts their ability to make financial decisions. We also acknowledge though that not all is equal when it comes to retirement and financial decision making (e.g. differing life expectancy rates between Indigenous and non-Indigenous peoples and access to wealth).

An individual's ability to make effective financial decisions is also impacted by social, economic and psychological factors (Huston, 2010), gender (Pinto & Coulson, 2011) and culture (Blue, 2016; Lahn, 2008; Pinto & Chan, 2010). Being financially literate and having the financial means to apply financial knowledge differs between individuals (Blue, 2016). For instance, an individual living on a low income who is financially literate will be constrained by their income when it comes to making effective financial decisions.

To determine an individual's financial literacy, there are three main areas that are tested. These questions are referred to as the Big Three: (1) the ability to calculate interest; (2) understanding the impact of inflation; and (3) the ability to understand risk diversification (Lusardi, 2019; Lusardi & Mitchell, 2011). Risk diversification now includes alternative financial products, as “the exponential growth in financial technology (fintech) is revolutionizing the way people make payments, decide about their financial investments, and seek financial advice” (Lusardi, 2019, p. 1).

Alternative financial products and services include financial tools that sit outside mainstream financial institutions. Two examples of alternative financial products are cryptocurrencies (or crypto assets¹) and NFTs; and buy now pay later is an example of an alternative financial service. Financial education on alternative financial products and services from an independent and trusted source is needed as most of the knowledge about these alternative tools comes from social media (Blue, Xing, & Pham, 2024; Blue, Xing, Pham, et al., 2024; Delfabbro et al., 2021). Many cryptocurrency owners also report making a purchase after speaking to family and/or friends (Financial Conduct Authority [FCA], 2019). Once cryptocurrencies are

acquired, handling them is reported to be quite complex (Steinmetz et al., 2021), leaving users vulnerable.

2.3 | External factor—Storage of cryptocurrencies and/or NFTs

Storing your cryptocurrencies and/or NFTs involves both internal and external factors. The internal factor is connected to an individual's information technology (IT) skills, and external factors are connected to the product and/or service available for storage. Safely securing crypto assets and NFTs is not always as straightforward as it may seem. There are risks with storing both on an exchange, as was recently reported with the downfall of FTX, which was one of the largest crypto exchanges in the world that collapsed in early November 2022 (Huang, 2022).

For any investors who had left their investments on the FTX exchange, they no longer had access to their collective \$8 billion that was lost (Chow, 2022). A lack of access occurred because FTX removed trading and withdrawal functionality from its exchange, leaving consumers who had stored their investments on the exchange unable to access their crypto assets (Sigalos, 2022).

Securing crypto assets and NFTs off the exchange comes with its own challenges. Voskoboynikov et al. (2020) report that:

Some participants considered keeping the private key private to avoid losses in potential shutdowns of exchanges. This, however, can only be done if the respective user understands the value of the private key. Some of our participants reported having accidentally deleted wallet files, while others did not understand what private and public keys were in the first place.

(p. 608)

Storing crypto assets on hard or cold wallets means storing them offline. This can be done using cold storage wallets such as Ledger, Trezor and Ellipal and requires the consumer to use a computer, download the software, record the seed phrase (also referred to as a recovery phrase), store that securely and create a pin to access the cold wallet in the future (Pitt, 2022). Access to the crypto assets will require the seed phrase and cold wallet. If the cold wallet is lost, the seed phrase can be used to swap the lost cold wallet for a new cold wallet. A private key is also required to access crypto assets (Moreland, 2022), which is not the seed phrase, although they are closely related. Private keys² and public keys have been described in this way:

Managing a blockchain address using private and public keys works similarly to a mailbox. The mailman can find your location and insert letters through the letter slot using the mail address (public key). However, the only person that can *retrieve* the contents from the mailbox is the one who owns the mailbox key (private key).

(Moreland, 2022, para 4)

A public key can be shared with others to receive crypto, “but only you, the person with the private key, can actually access the blockchain address and the funds stored in it. This is what makes you the owner – making it the pivotal element of securing your blockchain assets” (Moreland, 2022, para 5). A blockchain³ is used to distribute the transaction associated with crypto assets and NFTs. However, if an individual accidentally deletes and/or loses their private key, there is no resource through consumer protection measures.

2.4 | External factor—Lack of consumer protection

Investor protection is a major concern when it comes to purchasing cryptocurrencies (Avdeychik & Capozzi, 2018). Crypto exchanges are not regulated, meaning there are no consumer protections afforded to customers who purchase crypto. Storing a crypto asset and/or NFT also carries risks depending on what type of wallet is used for storage. The options include a custodial wallet (on an exchange), a soft wallet (on the web) or a hard wallet (offline). Crypto wallets store private and public keys. “A private key is a unique individual password to your individual crypto wallet address. A public key then adds an extra layer of security and ensures that your wallet can't be hacked” (Danial et al., 2022, p. 29). Bickers (2020) states:

It seems the more you know about key management, the more aware you are of the risk in dealing with them. For some users, the alternative of managing the keys themselves may be more of a risk than allowing the exchange to do it for them.
(p. 179)

Aside of the risk associated with storage, there are also tax implications, crypto owners must consider as cryptocurrencies and NFTs are treated as assets in Australia.

2.5 | External factor—Tax consequences

Since the rise in finance apps and opportunities to invest online, Australians may not be fully aware of the potential tax consequences and may experience adverse events, which leave them financially vulnerable. For example, buying and selling cryptocurrencies may cause a capital gains tax or capital loss tax event if it is sold for more or for less than its initial cost (Australian Government, n.d.). And each crypto asset triggers its own tax event. If investors are not aware, they may omit this on their annual tax return and be contacted by the Australian Taxation Office later when it is discovered.

A taxable event occurs because cryptocurrencies are treated as an asset in Australia rather than a currency. Cryptocurrencies are digital assets that can be used as a medium of exchange and/or held as an investment. They sit outside centralised digital currencies and central banking systems and are mostly decentralised and use distributed ledger technology (blockchain) (Chohan, 2017; Gipp et al., 2015; Othman et al., 2020). Anyone with a smartphone who is 16 years old and over can purchase cryptocurrencies from a cryptocurrency exchange. However, the individual is responsible for storing their crypto asset and as mentioned previously, has no recourse if they lose their purchase either on the exchange and/or via holding their own private keys.

In the next section, we share the theoretical framework and research questions that guided this study.

3 | THEORETICAL FRAMEWORK

We have taken insights from sociological perspectives of vulnerabilities (Forbes-Mewett & Nguyen-Trung, 2020; Gentry et al., 1994; Hill, 2001; Hill & Stamey, 1990; Menzel Baker et al., 2005; Penaloza, 1995; Smith & Copper-Martin, 1997), which point to internal and external conditions or factors that impact vulnerability. Forbes-Mewett and Nguyen-Trung (2020) state that from a sociological perspectives, vulnerability stems from “weaknesses of social systems” that will in combination with other external factors (e.g. “...distribution of resources; physical elements; logistical elements; and other environmental conditions such as economic,

social, and political upheaval and violence” Menzel Baker et al., 2005, p. 130) create conditions where being harmed is possible (p. 20). Gentry et al. (1994) define vulnerability based on who experiences it noting that Menzel Baker et al. (2005, p. 136) argue that “all of us, on occasion, will experience vulnerability”. Menzel Baker et al. (2005) define consumer vulnerability as the following:

Consumer vulnerability is a state of powerlessness that arises from an imbalance in marketplace interactions or from the consumption of marketing messages and products. It occurs when control is not in an individual's hands, creating a dependence on external factors (e.g., marketers) to create fairness in the marketplace. The actual vulnerability arises from the *interaction* of individual states, individual characteristics, and external conditions within a context where consumption goals may be hindered and the experience affects personal and social perceptions of self. (Menzel Baker et al., 2005, p. 134)

Noting that the above definition focusses on vulnerability in a consumer setting and “does not say who is vulnerable, because everyone has the potential to be” (p. 134).

With any financial decision, there are behavioural aspects that influence decision making. Cryptocurrency trading is reported to have links to behavioural addiction research (Delfabbro et al., 2021). They state that the reasons are related to the ability to buy and sell at any time and on any day and that cryptocurrencies are strongly influenced by social media, both external factors, which have the potential to create a fear of missing out (FOMO).

Purchasing cryptocurrencies are also deemed to be riskier than traditional investment trading, which closes at the end of the day, do not operate on the weekends or public holidays and in the case of sports are dependent games being played (Delfabbro et al., 2021). The internal factor of impulse control is needed to reduce the risk of gambling with crypto as the external factors, such as 24/7 exchange operating hours, are outside the control of the crypto owner.

Financial literacy levels are impacted by both internal and external factors. Internal factors include gender (women), Indigeneity, low education levels, lower incomes and English as a second language (Lusardi, 2019). External factors include access to financial education; social norms/cultural norms connected to personal finance; government policies and regulation; and advances in technology. Both internal and external factors relating to financial literacy and/or financial decision making may have an impact on an individual's vulnerability when making financial decisions. Later in the paper, we will analyse our findings under the lens of internal and external factors. Next, we share our data collection methods.

4 | DATA AND METHODS

Due to the rise in investment scams related to cryptocurrencies, we wanted to understand the vulnerabilities associated with buying and owning cryptocurrencies and NFTs. In this study, we sought to answer the following research questions: (1) Who is vulnerable when owning cryptocurrencies and/or NFTs? (2) What are the vulnerabilities (internal and external factors identified)? To answer these questions, we developed a survey, which was distributed online.

4.1 | Data collection

This research was carried out in 2022, and data collection was conducted after ethical clearance was obtained from the Queensland University of Technology where we were once all employed at. To investigate the sociodemographics of Australians investing in cryptocurrencies

and to identify any vulnerabilities, we designed the survey instrument (Appendix 1) based on the existing literature and knowledge about buying, storing and selling cryptocurrencies.

The survey instrument comprised of two screening questions (one about the participant's age and another confirming that they had purchased cryptocurrency before) and also included 40 questions (Appendix 1). The survey questions included some demographic information about the participants and information about their experience and interest in cryptocurrencies and/or NFTs and how they had learned about alternative finance products.

We also included three questions that focussed on their financial literacy and/or financial understanding connected to cryptocurrencies. These questions were deliberately included to gauge the participants financial literacy of cryptocurrencies. The questions included are as follows: whether participants were familiar with a common and important phrase: "not your keys, not your coins" (for Bitcoin owners); another question ask participants to calculate how much Bitcoin had increased by; and the third question ask participants whether they understood the tax consequences for selling cryptocurrencies (more details in the Results section of this paper).

The study was targeted at Australians who were 18 or older and had purchased cryptocurrencies and/or NFTs. We recruited 745 participants from Qualtrics who partners with over 20 online sample providers to supply a network of diverse quality participants who act as a client base. The participants were recruited anonymously online and randomly meaning that any participant in their database who was 18 or older and had purchased cryptocurrencies and/or NFTs had an equal chance of participating. Although participants could opt in to complete our survey, we did seek a mix of socioeconomic profiles, such as gender, ethnicity, socioeconomic status and education levels. Qualtrics managed the balance of demographic characteristics.

We remained outside the recruitment process as we paid Qualtrics to recruit participants based on our requirements. The number of participants we recruited was based on the budget available to conduct this research.

4.2 | Data analysis

This large dataset provided rich demographic information and an invaluable insight into Australians' financial penetration in cryptocurrencies and NFTs and their potential vulnerability. We conducted a preliminary descriptive analysis of the 745 survey responses using SPSS 28 to identify the sociodemographic features of the research sample, which we highlight in the next section.

Based on the summary statistics regarding the frequency and proportion of the participants' demographic information, chi-squared tests were further conducted to examine whether they differed significantly in gender, age range and educational level. The sections that follow reveal the demographic characteristics of cryptocurrency and NFT owners as well as the potential vulnerabilities.

5 | RESULTS AND DISCUSSION

5.1 | Who owns cryptocurrencies and/or NFTs?

It was a requirement that all participants ($n=745$) had purchased cryptocurrencies, and in addition, 140 participants (18.8 per cent) had also purchased NFTs. Of our sample of Australians who had purchased cryptocurrencies, female participants ($n=413$, 55.5 per cent) appeared to outnumber male participants ($n=325$, 43.6 per cent). Our sample differs from that previously reported (Steinmetz et al., 2021; Wciom, 2019) with a higher percentage of crypto owning females completing this survey.

Approximately half of the sample ($n=360$) were aged between 25 and 44 years. Among all the participants, only 8.3 per cent ($n=62$) were Aboriginal and Torres Strait Islander peoples. Most of the participants ($n=639$, 85.7 per cent) were employed, either undertaking full-time ($n=457$, 61.3 per cent) or part-time and/or casual jobs ($n=182$, 24.4 per cent), with the remaining 14 per cent ($n=104$) not working. The highest educational level varied from some high school, high school, through TAFE, to university, accounting for 3.2 per cent ($n=24$), 17.3 per cent ($n=129$), 22.4 per cent ($n=167$) and 55.7 per cent ($n=415$), respectively. Nearly half of the participants ($n=380$) owned their house, while 36.9 per cent of the sample ($n=274$) had mortgages on their houses (Table 1).

When examining the participants who purchased NFTs, as shown in Table 1, a larger proportion of participants ($n=111$, 15.3 per cent) were non-Indigenous than those who were Indigenous ($n=27$, 3.7 per cent), $\chi^2=26.36$, $p<.001$. This result mirrors the population distribution in Australia because there is a larger proportion of non-Indigenous people than Indigenous people living in Australia.

Educational-level differences were identified, $\chi^2=14.69$, $p<.001$, with a chi-squared test revealing that a larger proportion of participants with university degrees ($n=98$, 13.2 per cent) had purchased NFTs than those who had studied at TAFE ($n=18$, 2.4 per cent). Employment condition differences were also identified, $\chi^2=10.63$, $p=.005$, with a chi-squared test revealing

TABLE 1 Differences in purchasing cryptocurrencies and non-fungible tokens by demographic characteristics.

	Purchasing cryptos	Purchasing NFTs		
	Yes <i>N</i> (% of total)	Yes <i>N</i> (% of total)	No <i>N</i> (% of total)	χ^2
Gender				2.18, $p=.14$
Female	413 (55.4%)	70 (9.4%)	343 (46%)	
Male	325 (43.6%)	69 (9.3%)	256 (34.4%)	
Non-binary	6 (0.8%)	1 (0.1%)	5 (0.7%)	
Prefer not to say	1 (0.1%)	0 (0.0%)	1 (0.1%)	
Indigeneity				26.36, $p<.001$
Indigenous	62 (8.3%)	27 (3.7%)	35 (4.8%)	
Non-Indigenous	662 (89%)	111 (15.3%)	551 (76.11%)	
Prefer not to say	20 (2.7%)	2 (0.3%)	18 (2.4%)	
Educational level				14.69, $p<.001$
High school or below	153 (20.6%)	23 (3.1%)	130 (17.4%)	
TAFE	167 (22.4%)	18 (2.4%)	149 (20%)	
University	415 (55.7%)	98 (13.2%)	317 (42.6%)	
Other	10 (1.3%)	1 (0.1%)	9 (1.2%)	
Employment condition				10.63, $p=.005$
Full-time	457 (61.3%)	103 (13.8%)	354 (47.5%)	
Part-time/casual	182 (24.4%)	23 (3.1%)	159 (21.3%)	
No	104 (14.0%)	14 (1.9%)	90 (12.1%)	
Prefer not to say	2 (0.3%)	0 (0.0%)	2 (0.3%)	

Note: We did not perform the chi-square test for "purchasing cryptocurrencies," which is a constant because all participants purchased cryptocurrencies. When performing the chi-squared test for independence, some categories with lower frequencies were removed for meeting the statistical assumptions of chi-squared test, that is, the lowest expected frequency in any cell should be 5 or more (Pallant, 2020). The categories removed include: "Non-binary" and "Prefer not to say" from the "Gender" category; "Prefer not to say" from the "Indigeneity" category; "Other" from the "Educational level"; "Prefer not to say" from the "Employment condition". Values bolded denote significance was found.

that a significantly larger proportion of participants who were employed full-time ($n=457$, 61.3 per cent) had purchased NFTs than those who undertook part-time/casual jobs ($n=182$, 24.4 per cent) or were not employed ($n=104$, 14.0 per cent).

There were no significant associations with gender identified. However, the results show that participants who were non-Indigenous, or had received university education or were employed full-time were more likely to purchase NFTs. This suggests the internal factors including ethnicity, education and income influenced NFT ownership.

Research by Voskobochnikov et al. (2020, p. 609) reported that “negative beliefs about cryptocurrency use” are often the reason consumers do not own crypto and along with other misconceptions held by non-owners. We also know that NFTs and cryptocurrencies have a strong social media presence, which is an external factor, and may influence an individual's ability to control their impulses due to a FOMO. Next, we wanted to identify vulnerabilities that exist when purchasing crypto assets and NFTs and identify who is vulnerable.

5.2 | What are the vulnerabilities and who is vulnerable?

In this section, we report on the statistical analyses regarding the vulnerabilities reported by the participants and identify vulnerable groups by demographic information in relation to each vulnerability. We also link the findings back to internal and external factors and whether the findings support or counter the internal and/or external factors described in the existing literature.

5.3 | Concerns about storing crypto and NFTs

When surveyed about where crypto owners stored their cryptocurrencies, most of the participants ($n=443$, 59.5 per cent) reported using a software wallet, followed by 47.0 per cent ($n=350$) who used an exchange, and 16.9 per cent ($n=126$) who used a hardware wallet. Regarding the storage of NFTs, most participants ($n=86$, 11.5 per cent) used a software wallet, followed by 7.4 per cent ($n=55$) who used an exchange, and 5.8 per cent ($n=43$) who used a hardware wallet.

We further surveyed whether the participants had any concerns about the storage of cryptocurrencies or NFTs. Approximately 34 per cent of participants ($n=248$) had concerns about the storage of cryptocurrencies, and 36.2 per cent of participants ($n=50$) were concerned about storing NFTs. Among 248 participants who had concerns about the storage of cryptocurrencies, 62.1 per cent ($n=154$) stored their cryptocurrencies in a software wallet, followed by 50 per cent ($n=124$) who used an exchange, and 23.4 per cent ($n=58$) who used a hardware wallet. For the 50 participants who were concerned about the storage of NFTs, 34 participants (68.0 per cent) stored their NFTs in a software wallet, followed by 19 participants (38 per cent) who used a hardware wallet and 17 participants (34.0 per cent) who used an exchange.

Eighty-six participants further reported concerns over the security of cryptocurrencies, worrying about potential money losses while 26 participants also worried about being hacked and another 15 participants felt distressed about potential theft and fraud due to cryptocurrencies, along with three participants who feared their privacy disclosure. Apart from the concerns above, three participants had doubts about the legitimacy of owning cryptocurrencies or NFTs and seven participants were not confident about their knowledge of cryptocurrencies or NFTs.

Both storage and security of cryptocurrencies and NFTs are external factors. However, storage also involves the internal factor of IT literacy relating to cryptocurrencies and NFTs. We also analysed the education levels, which is an internal factor against the external factor of storage.

Among the participants, educational-level differences were identified, $\chi^2 = 17.77$, $p < .001$, with chi-squared tests revealing that a smaller proportion of those who had concerns about the storage of cryptocurrencies had graduated from high school or below ($n = 33$, 4.4 per cent) compared with those who had attended TAFE ($n = 49$, 6.6 per cent) or university degree ($n = 163$, 21.9 per cent). Participants' concerns about the storage of cryptocurrencies did not differ significantly by gender, age range, Indigeneity and employment condition. However, from these results, it is suggested that participants with a university degree or TAFE certificate tend to have concerns over the storage of cryptocurrencies.

Internal factors such as education levels impact an individual's awareness of the associated storage risks of buying cryptocurrencies. Thus, lower education levels could make an individual more vulnerable when purchasing and storing cryptocurrencies, which has been previously reported in the literature regarding financial literacy levels (Lusardi, 2019). There were no significant associations with demographic characteristics of the participants who had concerns about the storage of NFTs.

We further examined the relationship between the participants who had lost funds and had concerns over the storage of cryptocurrencies; a larger proportion of participants who had concerns about the security of cryptocurrencies had lost their investments ($n = 50$, 7.1 per cent) than those who did not have concerns ($n = 19$, 2.7 per cent), $\chi^2 = 47.97$, $p < .001^*$. Analysis of the results suggest, on the one hand, that a certain number of the participants were concerned about the storage issues of cryptocurrencies or NFTs, revealing that they felt vulnerable in securing their cryptocurrencies or NFTs.

Most participants had no concerns about the security of cryptocurrencies or NFTs, which is worrisome because 50 per cent of participants store their crypto assets on an exchange. This finding suggests that a lack of awareness about the potential to lose crypto assets stored on exchanges, which is connected to IT literacy.

To determine whether a lack of concern about the storage of crypto assets and NFTs was an issue, we further investigated whether there was any relationship between the storage of cryptocurrencies or NFTs and storage on the exchange. Of 248 participants who had concerns over the storage of cryptocurrencies, 124 participants (50 per cent) stored their cryptocurrencies on the exchange. A chi-squared test indicated no significant association between participants' concern over the storage of cryptocurrencies and the storage of cryptocurrencies on the exchange, $\chi^2 = .67$, $p = .462$. Of the 50 participants who had concerns over the security of NFTs, 17 participants (34.0%) stored their NFTs on an exchange, while a chi-squared test indicated no significant association between participants' concerns over the storage of NFTs and the storage of NFTs on an exchange, $\chi^2 = 1.12$, $p = .366$.

It can be concluded that there is no significant relationship between concerns over storage of cryptocurrencies or NFTs and storage on an exchange. These findings reveal that higher risk-taking behaviour is often associated with cryptocurrency owners (Delfabbro et al., 2021). For example, leaving cryptocurrencies on an exchange means that the owner is willing to risk losing their investment if anything happens to the exchange such as it being hacked.

5.4 | Loss due to storage issues

Participants were also surveyed about whether they had lost their cryptocurrencies or NFTs due to storage issues. Sixty-nine participants (9.4 per cent) reported that they lost their cryptocurrencies due to storage issues, and 11 participants (7.9 per cent) lost their NFTs to poor storage. Overall, only a minority of participants reported that they lost their cryptocurrencies or NFTs because of inappropriate storage. Among those who reported the loss of cryptocurrencies, almost 55.1 per cent ($n = 31$) had lost over \$1000. Of those who reported the loss of NFTs, seven participants (63.6 per cent) had lost <\$1000, and only four participants (36.4 per cent)

had lost over \$1000. Storage of crypto and/or NFTs continues to be reported as a vulnerability as there are often reports about crypto investors losing their investments due to insolvent exchanges (see FTX, mentioned earlier).

When examining the participants who lost funds due to cryptocurrencies, as shown in Table 2, a larger proportion of male participants ($n=38$, 5.3 per cent) lost funds than females ($n=30$, 4.2 per cent), $\chi^2=4.43$, $p=.040$. Similarly, a larger proportion of Indigenous participants ($n=16$, 2.2 per cent) lost money than non-Indigenous participants ($n=50$, 6.9 per cent), $\chi^2=26.67$, $p<.001$. Housing condition differences were identified, $\chi^2=6.44$, $p=.040$, with a chi-squared test revealing that a larger portion of participants who owned a home with a mortgage ($n=32$, 4.5 per cent) lost funds than those who owned a home without a mortgage ($n=5$, 2.1 per cent) or participants who rent houses ($n=21$, 2.9 per cent). A larger proportion of participants who lived with others ($n=518$, 71.8%) are less likely to lose funds than those who lived on their own ($n=132$, 18.3 per cent), $\chi^2=16.94$, $p<.001$.

Internal factors including gender (i.e. male), race (i.e. Indigenous), home ownership (with a mortgage) and living with others all contributed to cryptocurrency owners losing their investments. Previous research by Delfabbro et al. (2021) aligned gambling and addictions with cryptocurrency ownership, which may warrant further exploration based on the demographics of crypto owners found in our study.

Regarding the loss of NFTs due to storage issues, a larger proportion of Indigenous participants ($n=6$, 4.4%) lost funds than non-Indigenous participants ($n=5$, 3.7 per cent), $\chi^2=8.81$, $p=.009$. The results indicate that males, Indigenous peoples, mortgage holders or participants who lived on their own were more likely to lose funds due to cryptocurrencies while Indigenous participants were also more likely to lose funds due to NFTs.

The finding that Indigenous peoples were both more likely to lose both cryptocurrency assets and NFTs is of concern. The above-mentioned finding may be connected to the external vulnerability factor associated with a lack of strong social networks (Salignac et al., 2020) in the alternative financial investment space, which has been shown to increase vulnerability.

6 | UNSOLICITED ADVICE AND LIMITED CHANNELS FOR LEARNING ABOUT CRYPTOCURRENCIES AND NFTs

6.1 | Unsolicited advice

In this survey, 28.3 per cent of participants ($n=210$) reported that they had received unsolicited investment advice relating to cryptocurrencies. The participants stated that they were mostly approached by e-mail ($n=108$, 51.4 per cent), direct message ($n=79$, 37.6 per cent), followed by phone ($n=67$, 31.9 per cent) and text message ($n=45$, 21.4%). A minority of participants ($n=15$, 7.1 per cent) also indicated that they were approached on social media such as Instagram, Facebook and YouTube, or in person.

When examining the participants who had received unsolicited investment advice relating to cryptocurrencies, a larger proportion of participants self-identified as non-Indigenous ($n=175$, 23.5 per cent) than those who self-identified as Indigenous ($n=30$, 4.0 per cent), $\chi^2=13.72$, $p<.001$. Interestingly, non-Indigenous Australians were more likely to receive unsolicited cryptocurrency investment advice.

There were no significant associations with internal factor such as gender, age range, Indigeneity and employment conditions, thus showing despite non-Indigenous peoples being less likely to receive unsolicited investment advice, anyone can be vulnerable when it comes to cryptocurrency and NFT investment scams. This finding aligns with the increase in scams operating in Australia and targeting all (Scamwatch, 2021).

TABLE 2 Demographic information of survey participants who lost funds.

Variable	Category	Loss of cryptocurrencies			Loss of NFTs		
		Yes % of total	No % of total	<i>p</i> Value	Yes % of total	No % of total	<i>p</i> Value
Gender	Male	38 (5.3%)	275 (38.1%)	<i>p</i> = .040*	6 (4.4%)	60 (44.1%)	<i>p</i> = .761*
	Female	30 (4.2%)	371 (51.5%)		5 (3.7%)	64 (47.1%)	
	Non-binary	1 (0.1%)	5 (0.7%)		0 (0.0%)	1 (0.7%)	
	Prefer not to say	0 (0.0%)	1 (0.1%)		0 (0.0%)	0 (0.0%)	
Indigeneity	Indigenous	16 (2.2%)	43 (8.3%)	<i>p</i> < .001*	6 (4.4%)	21 (15.4%)	<i>p</i> = .009*
	Non-Indigenous	50 (6.9%)	592 (82.2%)		5 (3.7%)	102 (75.0%)	
	Prefer not to say	2 (0.3%)	17 (2.4%)		0 (0.0%)	2 (1.5%)	
Housing condition				6.44, <i>p</i> = .040			N/A
	Own home with a mortgage	32 (4.5%)	234 (32.5%)		7 (5.2%)	52 (38.5%)	
	Own home without a mortgage	15 (2.1%)	86 (12.0%)		3 (2.2%)	23 (17.0%)	
	Rent your home	21 (2.9%)	275 (38.2%)		1 (0.7%)	42 (31.1%)	
	Other	1 (0.1%)	22 (3.1%)		0 (0.0%)	3 (2.2%)	
	Prefer not to say	0 (0.0%)	33 (4.6%)		0 (0.0%)	4 (3.0%)	
Living condition				<i>p</i> < .001*			N/A
	Living on my own	29 (4.0%)	131 (18.3%)		5 (3.7%)	28 (20.6%)	
	Living with others	40 (5.5%)	518 (71.8%)		6 (4.4%)	96 (70.6%)	
	Prefer not to say	0 (0.0%)	2 (0.3%)		0 (0.0%)	1 (0.7%)	

Note: The categories removed include: “Non-binary” and “Prefer not to say” from “Gender” category; “Prefer not to say” from the “Indigeneity” category; “Other” and “Prefer not to say” from “Housing condition”; and “Prefer not to say” from “Living condition”. *This refers to the result from Fisher's exact test when the expected cell count was too small for a chi-square test. N/A means this category failed to meet the statistical assumptions of a chi-square test. Values bolded denote significance was found.

6.2 | All that hype—Reasons behind crypto purchases

We also investigated the reasons participants buy or hold cryptocurrencies or NFTs. It was found that most participants ($n=623$, 83.6 per cent) reported that they purchased crypto assets and NFTs for potential investment (Figure 1). Another 54.1 per cent of the participants purchased crypto due to the long-term value of cryptocurrencies, followed by 14.1 per cent seeking investment in decentralised currencies and 5.6 per cent for purchasing NFTs. As for NFT owners, they sought a potential investment (11 per cent) and a long-term store of value (11.8 per cent). Notably, a relatively large number of participants reported that they were affected by “All the hype on social media” when they bought cryptocurrencies ($n=223$, 29.9 per cent) and NFTs ($n=50$, 35.7 per cent).

Table 3 reports the demographic characteristics of the participants who purchased cryptocurrencies due to “All the hype on social media”. Chi-squared test results indicate that a larger proportion of participants ($n=139$, 18.7 per cent) were female ($n=81$, 10.9 per cent), $\chi^2=6.63$, $p=.012$. Similarly, a larger proportion of participants ($n=25$, 3.5 per cent) were non-Indigenous than those who were Indigenous ($n=193$, 26.7 per cent), $\chi^2=3.36$, $p=.082$. Educational-level differences were identified, $\chi^2=8.64$, $p=.013$, with a chi-squared test revealing a larger proportion of participants with a university degree ($n=138$, 18.8 per cent) than those who had undertaken TAFE courses ($n=35$, 4.8%).

The results suggest that female participants, non-Indigenous participants or participants with a tertiary degree were more likely to buy cryptocurrencies due to “All the hype on social media.” Employment condition differences were identified, $\chi^2=6.40$, $p=.041$, with chi-squared test revealing that a significantly larger proportion of participants who were employed full-time ($n=39$, 5.2 per cent) were buying NFTs than those who undertook part-time/casual jobs ($n=8$, 1.1 per cent) or were unemployed ($n=3$, 0.4 per cent). This means that participants who were employed full-time were more likely to buy NFTs based on “All the hype on social media.”

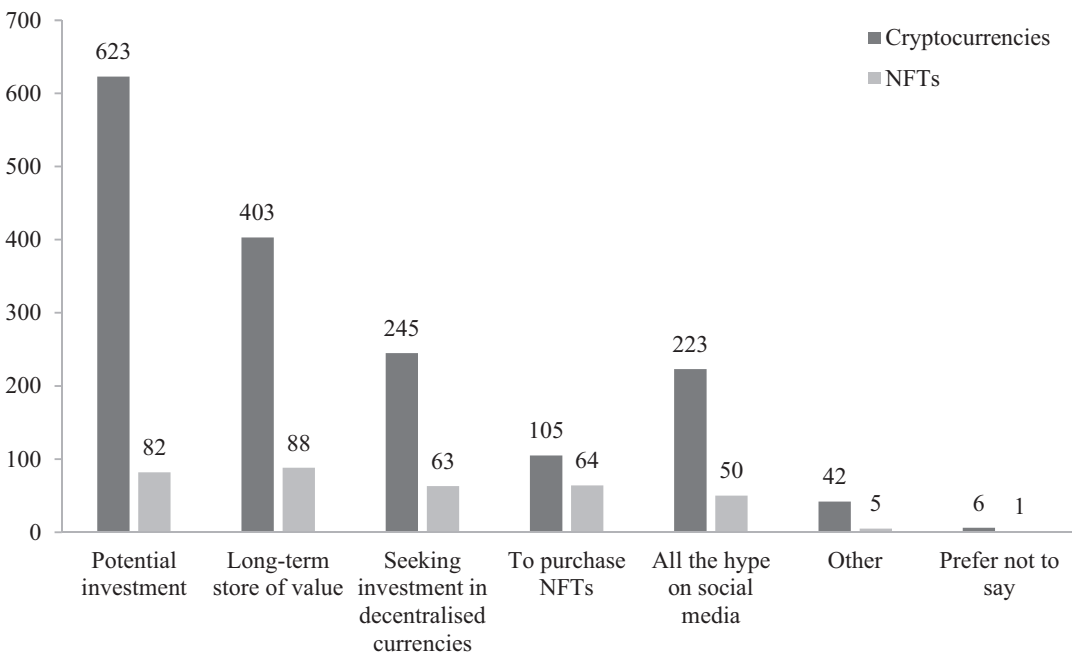


FIGURE 1 Reasons for buying cryptocurrencies or non-fungible tokens.

TABLE 3 Demographic information of survey participants who selected “All the hype of social media.”

Variable	Category	Cryptocurrencies			NFTs		
		Yes % of total	No % of total	Chi-square	Yes % of total	No % of total	Chi-square
Gender	Male	81 (10.9%)	244 (32.8%)	$p = .012^*$	17 (2.3%)	308 (41.3%)	$p = .183^*$
	Female	139 (18.7%)	274 (36.8%)		32 (4.3%)	381 (51.1%)	
	Non-binary	3 (0.4%)	3 (0.4%)		1 (0.1%)	5 (0.7%)	
	Prefer not to say	0 (0.0%)	1 (0.1%)		0 (0.0%)	1 (0.1%)	
Indigeneity	Indigenous	25 (3.5%)	37 (5.1%)	$p = .082^*$	14 (1.9%)	48 (6.6%)	N/A
	Non-Indigenous	193 (26.7%)	469 (64.8%)		35 (4.8%)	627 (86.6%)	
	Prefer not to say	4 (0.5%)	16 (2.2%)		1 (0.1%)	19 (2.6%)	
Employment condition	Full-time	141 (18.9%)	316 (42.4%)	0.56, $p = .756$	39 (5.2%)	418 (56.1%)	6.4, $p = .041$
	Part-time and/or casual	51 (6.8%)	131 (17.6%)		8 (1.1%)	174 (23.4%)	
	Not working	30 (4.0%)	74 (9.9%)		3 (0.4%)	101 (13.6%)	
	Prefer not to say	1 (0.1%)	1 (0.1%)		0 (0.0%)	2 (0.3%)	
Educational level	High school or below	47 (6.4%)	106 (14.4%)	8.64, $p = .013$	8 (1.1%)	145 (19.7%)	5.61, $p = .061$
	TAFE	35 (4.8%)	132 (18.0%)		6 (0.8%)	161 (21.9%)	
	University	138 (18.8%)	277 (37.7%)		36 (4.9%)	379 (51.6%)	
	Other	3 (0.4%)	7 (0.9%)		0 (0.0%)	10 (1.3%)	

Note: The categories removed include: “Non-binary” and “Prefer not to say” from the Gender category; “Prefer not to say” from the Indigeneity category; “Other” from the Educational level degree; and “Prefer not to say” from the Employment condition category. *This refers to the result from Fisher’s exact test when the expected cell count was too small for chi-square tests. N/A means this category failed to meet the statistical assumptions of a chi-square test. Values bolded denote significance was found.

The internal factor of income appears to have influenced purchasing NFTs along with the external factor of social media influencers. It is not surprising that individuals with income are buying alternative investments that are promoted on social media. However, what is surprising is that females appear to be the most influenced by social media. Young adults have been previously found (Blue, Xing, & Pham, 2024) to be strongly influenced by social media as that was their main source of learning about cryptocurrencies and NFTs.

6.3 | Limited ways to learn about cryptocurrencies and NFTs

In this survey, we also asked about how participants learned about cryptocurrencies or NFTs. Most of the participants learned from social media about cryptocurrencies ($n=452$, 60.7 per cent) and NFTs ($n=85$, 60.7 per cent) (Figure 2). Family or friends also served a key role, occupying 60.7 per cent ($n=452$) for cryptocurrencies and 57.1 per cent ($n=60$) for NFTs. Financial planners and advisors were also an important source for participants to acquire information about cryptocurrencies ($n=109$, 14.6 per cent) and NFTs ($n=26$, 18.6 per cent). A minority of participants obtained information from their school, TAFE or university about cryptocurrencies ($n=43$, 5.8 per cent) and NFTs ($n=27$, 19.3 per cent).

According to the above findings, we can infer that the participants in this survey were seeking knowledge about cryptocurrencies and NFTs that they did not have, which is an internal factor. The findings also reveal that they had rather limited channels of learning about cryptocurrencies and NFTs, which is an external factor. Our results confirm the findings from Blue, Xing, and Pham (2024) and Blue, Xing, Pham, et al. (2024) that social media influencers continue to be the source of truth when it comes to alternative financial products. Understanding that individuals can become vulnerable through social media influencers highlights the need for more education in the space to critically unpack social media influencer (often referred to as a finfluencer because their message is about personal finance) messaging especially for young adults.

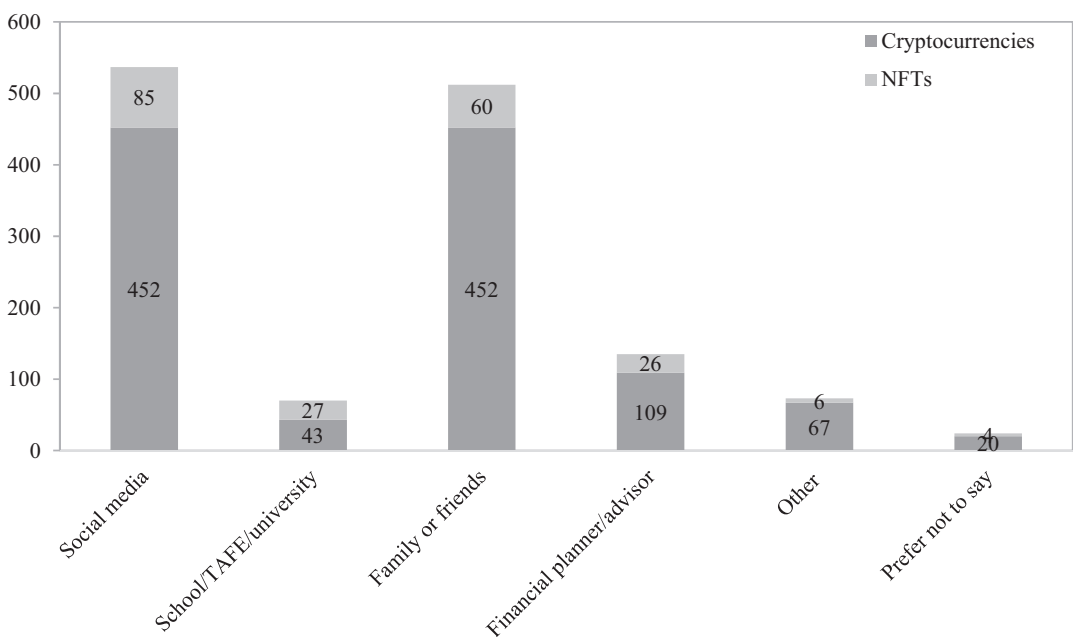


FIGURE 2 Sources for learning about cryptocurrencies and non-fungible tokens.

7 | INSUFFICIENT FINANCIAL LITERACY AND IT LITERACY IN CRYPTOCURRENCIES

7.1 | Insufficient financial literacy in cryptocurrencies

In this study, we measured participants' financial literacy in cryptocurrencies with three questions. The first question (Q1) asked whether participants who only bought Bitcoin ($n=263$, 35.3%) believed the saying “not your keys, not your coins⁴,” and it was found that most of the participants ($n=582$, 78.1%) tended to buy Bitcoin. We found that 28.5 per cent of participants ($n=75$) agreed with the saying “not your keys, not your coins,” while 20.9 per cent of the

TABLE 4 Demographic information of survey participants for Q1.

Variable	Category	Q1			Chi-square
		Correct % of total	False % of total	Not sure % of total	
Gender					34.96, $p < .001$
	Male	138 (18.6%)	59 (8.0%)	126 (17.0%)	
	Female	97 (13.1%)	72 (9.7%)	242 (32.7%)	
	Non-binary	2 (0.3%)	0 (0.0%)	4 (0.5%)	
Indigeneity	Prefer not to say	0 (0.0%)	1 (0.1%)	0 (0.0%)	15.61, $p < .001$
	Indigenous	32 (4.3%)	12 (1.6%)	17 (2.3%)	
	Non-Indigenous	199 (26.9%)	113 (15.3%)	347 (46.9%)	
	Prefer not to say	5 (0.7%)	7 (0.9%)	8 (1.1%)	
Employment condition					20.98, $p < .001$
	Full-time	166 (22.4%)	89 (12.0%)	198 (26.4%)	
	Part-time/casual	42 (5.7%)	26 (3.5%)	114 (15.4%)	
	No	28 (3.8%)	17 (2.3%)	59 (8.0%)	
Housing condition	Prefer not to say	1 (0.1%)	0 (0.0%)	1 (0.1%)	14.85, $p = .005$
	Own home with a mortgage	85 (11.5%)	59 (8.0%)	129 (17.5%)	
	Own home without a mortgage	46 (6.2%)	21 (2.8%)	39 (5.3%)	
	Rent home	89 (12.0%)	45 (6.1%)	169 (22.9%)	
	Other	5 (0.7%)	3 (0.4%)	15 (2.0%)	
Living condition	Prefer not to say	11 (1.5%)	3 (0.4%)	20 (2.7%)	9.24, $p = .01$
	Living on own	68 (9.2%)	29 (3.9%)	68 (9.2%)	
	Living with others	168 (22.7%)	102 (13.8%)	304 (41.0%)	
	Prefer not to say	1 (0.1%)	1 (0.1%)	0 (0.0%)	
Language at home other than English					12.72, $p = .002$
	Yes	46 (6.2%)	14 (1.9%)	37 (5.0%)	
	No	186 (25.2%)	118 (16.0%)	330 (44.7%)	
	Prefer not to say	3 (0.4%)	0 (0.0%)	5 (0.7%)	

Note: The categories removed include: “Non-binary” and “Prefer not to say” from the Gender category; “Prefer not to say” from the Indigeneity category; “Prefer not to say” from Employment condition; and “Prefer not to say” from Living condition; “Other” and “Prefer not to say” from Housing condition; and “Prefer not to say” from the Language at home other than English category”. Values bolded denote was found significance.

participants ($n=55$) rejected the saying. Approximately half of the participants ($n=133$, 50.6%) were feeling unsure about this saying. We further examined the demographic features of participants who offered different answers to Q1 (Table 4).

As shown in Table 4, a chi-squared test indicated a larger proportion of male participants ($n=138$, 18.8 per cent) answered Q1 correctly than the female participants ($n=97$, 13.2 per cent), $\chi^2=34.96$, $p<.001$. Similarly, a larger proportion of non-Indigenous participants ($n=347$, 46.9%) were unsure about this saying than those who were Indigenous ($n=17$, 2.3 per cent), $\chi^2=15.61$, $p=.082$.

Housing condition differences were identified, $\chi^2=14.85$, $p=.005$, with a chi-squared test revealing that a smaller portion of participants who owned a home without a mortgage ($n=39$, 5.3 per cent) were unsure about this saying than those who rented houses ($n=169$, 24.8 per cent). Living condition differences were also identified, $\chi^2=6.31$, $p=.01$, with a chi-squared test revealing that a larger proportion of participants who lived on their own ($n=68$, 9.2%) were unsure of this saying than those who lived with others ($n=304$, 41.0 per cent).

Differences regarding “Language at home other than English” were identified, with chi-squared tests revealing that a larger proportion of participants who spoke another language at home ($n=330$, 44.7 per cent) were unsure of this saying than those who spoke English at home ($n=37$, 5.0 per cent).

Taken together, female participants, non-Indigenous participants, participants who rented their home, participants who lived on their own or participants who spoke a language at home other than English were more likely to have limited knowledge about Q1. These are all internal factors that could contribute to vulnerabilities when purchasing cryptocurrencies when limited IT literacy is known about storage.

The second question (Q2) required the participant to calculate interest relating to Bitcoin—“If Bitcoin was worth \$40,000 when you purchased and 12 months later it was worth \$80,000, by how much has Bitcoin increased?” More than half the participants ($n=426$, 57.2 per cent) chose the correct answer, while the remaining participants were either unsure ($n=64$, 8.6 per cent) or chose the wrong answer ($n=232$, 31.3 per cent), and 2.8 per cent of participants ($n=21$) chose “prefer not to say.”

Table 5 displays the cross-tabulation analysis of the demographic information of participants who responded to Q2. Chi-squared results indicated a larger proportion of male participants ($n=215$, 29.8 per cent) answered correctly when compared to females ($n=206$, 28.5 per cent), $\chi^2=26.31$, $p<.001$. The differences regarding “Language at home other than English”

TABLE 5 Demographic information of survey participants for Q2.

Variable	Category	Q2			Chi-square
		Correct % of total	False % of total	Not sure % of total	
Gender	Male	215 (29.8%)	70 (9.7%)	26 (3.6%)	26.31, $p<.001$
	Female	206 (28.5%)	161 (22.3%)	37 (5.1%)	
	Non-binary	5 (0.7%)	1 (0.1%)	0 (0.0%)	
	Prefer not to say	0 (0.0%)	0 (0.0%)	1 (0.1%)	
Language at home other than English	Yes	66 (9.2%)	20 (2.8%)	9 (1.3%)	6.31, $p=.043$
	No	356 (49.4%)	210 (29.2%)	53 (7.4%)	
	Prefer not to say	4 (0.6%)	0 (0.0%)	2 (0.3%)	

Note: The categories removed include: “Non-binary” and “Prefer not to say” from the Gender category; and “Prefer not to say” from the Language at home other than English category. Values bolded denote was found significance.

were also identified, with chi-squared tests revealing that a larger proportion of participants who spoke another language at home ($n=210$, 29.2 per cent) were incorrect than those who spoke English at home ($n=20$, 2.8 per cent).

The above results for Q2 suggest that female participants, or participants who spoke another language at home, were more likely to have insufficient knowledge. This question (Q2) required the participant to do some mathematical calculations, which points to a need for increased mathematical skills.

The third question (Q3) was designed to identify whether participants knew that selling cryptocurrencies can result in a taxable event. The analysis showed that 66 per cent of participants ($n=493$) believed so, but 8 per cent of the participants ($n=61$) perceived it was wrong and a quarter of the participants ($n=185$) were unsure about it. This means that nearly 30 per cent of participants were unsure about the tax consequences of buying and selling crypto assets.

Table 6 presents the demographic features of participants who answered Q3. A chi-squared test indicated a smaller proportion of male participants ($n=86$, 11.6 per cent) answered incorrectly than females ($n=162$, 21.8 per cent), $\chi^2=13.08$, $p<.001$. Employment condition differences were identified, $\chi^2=16.68$, $p<.001$, with a chi-squared test revealing that a significantly larger proportion of participants who were employed full-time ($n=327$, 44.0 per cent) were correct than those who undertook part-time or casual jobs ($n=107$, 14.4%) or were unemployed ($n=57$, 7.7 per cent). Similarly, educational-level differences were also identified, $\chi^2=7.89$, $p=.019$, with a chi-squared test revealing that a larger proportion of participants with a university degree ($n=290$, 39.0 per cent) were correct than those who had completed high school or below ($n=88$, 11.8 per cent). These results suggest that female participants, part-time employed

TABLE 6 Demographic information of survey participants for Q3.

Variable	Category	Correct		False		Chi-square
		Frequency	Per cent % of total	Frequency	Per cent % of total	
Gender						13.08, $p<.001$
	Male	238	32.0	86	11.6	
	Female	251	33.7	162	21.8	
	Non-binary	3	0.4	3	0.4	
	Prefer not to say	1	0.1	0	0	
Employment condition						16.68, $p<.001$
	Full-time	327	44.0	129	17.3	
	Part-time and/or casual	107	14.4	75	10.1	
	Not working	57	7.7	47	6.3	
	Prefer not to say	2	0.3	0	0	
Educational level						7.89, $p=.019$
	High school or below	88	11.8	65	8.7	
	TAFE	111	14.9	56	7.5	
	University	290	39.0	124	16.7	
	Other	4	0.5	6	0.8	

Note: The categories removed include: "Non-binary" and "Prefer not to say" from the Gender category; "Other" from the Educational level degree category; and "Prefer not to say" from the Employment condition category. Values bolded denote was found significance.

or participants without a job, or participants with high school or below were more likely to have insufficient knowledge of Q3.

In summary, the results from the three questions provide important insights, particularly that female participants and non-university degree participants may have insufficient financial literacy in cryptocurrencies in comparison with other groups. Internal factors such as gender and education levels seem to have played a role in the financial literacy knowledge associated with cryptocurrencies. Lusardi (2019) has previously reported that women and individuals with lower education levels have low financial literacy levels.

7.2 | Insufficient IT literacy in cryptocurrencies or NFTs

We also surveyed participants' self-rated technology skills for dealing with cryptocurrencies and NFTs. Most participants rated their technologies for purchasing, selling and storing cryptocurrencies or NFTs at average or above. For example, 52 per cent of participants ($n=386$) perceived their technology skills in cryptocurrencies as average, while there was a minority who rated their technology skills as below-average, with 8.3 per cent ($n=62$) for cryptocurrencies and 2.7 per cent ($n=20$) for NFTs. When examining the demographic features of participants and their self-rated levels of technical skills, a chi-squared test indicated a larger proportion of participants who had full-time jobs were more likely to rate their technology skills as average or above ($n=426$, 57.7 per cent) than those who were not working ($n=86$, 11.7 per cent), $\chi^2=13.91$, $p<.001$. This means that participants who were employed full-time tended to be more confident about their technology skills in cryptocurrencies or NFTs.

When comparing the participants who had lost their investments and who were worried about security and rated themselves low with technology skills, a three-way loglinear analysis was conducted. It was found that a larger proportion of participants who rated their technology skills as average or above were less likely to have concerns about the security of cryptocurrencies ($n=449$, 62.3%) than those who had self-rated as below-average technology skills ($n=27$, 3.7%), $\chi^2=15.27$, $p<.001$. It can thus be suggested that participants who were confident about their technology skills in cryptocurrencies were less concerned about the security of cryptocurrencies.

The association between the investment loss and the self-rated technology skills revealed that a larger proportion of participants who rated their technology skills of cryptocurrencies as average or above ($n=449$, 62.3%) were less likely to lose their investments than those who had self-rated below-average technology skills ($n=27$, 3.7%), $\chi^2=21.59$, $p<.001^*$.

The above-mentioned finding indicates that when participants rated their technology skills in cryptocurrencies as average or above, they were less likely to lose their investment than those who rated their skills as below-average. Internal factors including skills and knowledge about cryptocurrencies appear to have had an influence on the security of the crypto investments. Our finding is not surprising when comparing it to what has been shown in the financial literacy literature.

8 | SUMMARY OF RESULTS

In summary, our statistical analyses using cross-tabulations, chi-squared tests and loglinear analysis produced the following results. First, regarding who is purchasing cryptocurrencies, we found no significant difference associated with gender despite our sample having more female participants. With regard to NFTs, non-Indigenous Australians, or university-educated, or full-time employed people were more likely to purchase NFTs.

Second, we conclude from the analyses that the vulnerabilities encountered by crypto or NFTs owners included internal and external factors including the following: insufficient financial literacy and IT literacy in cryptocurrencies or NFTs; concerns over the security of cryptocurrencies or NFTs; receiving unsolicited advice; and limited channels of learning about cryptocurrencies and NFTs.

Third, as for who is vulnerable when purchasing or owning cryptocurrencies or NFTs, we found there to be two distinct groups. One group of participants possessed some features of socioeconomic disadvantage, including being female, identifying as Indigenous, having a high school or below education, working part-time or casual, renting a home or speaking a language other than English at home.

The above-mentioned participants were also vulnerable because they were influenced by social media hype and lacked sufficient financial literacy or IT literacy in cryptocurrencies or NFTs, or they had lost funds due to investments in cryptocurrencies or NFTs. Females showed limited knowledge on all three questions we asked relating to their cryptocurrency literacy. The demographics identified above align with the demographic characteristics found among individuals with lower financial literacy levels (Lusardi, 2019).

In contrast, the other vulnerable group is from a socioeconomic advantaged background, who were non-Indigenous, university-educated, employed full-time or owned their home with a mortgage. This group of well-educated, capital-rich participants also had sufficient financial literacy and IT literacy in cryptocurrencies or NFTs. They may assume that they would not be the “victim” of crypto investment scams, whereas such over-confidence or over-ambition may instead expose this group to vulnerabilities and risk. Over-confidence has also been previously shown in the financial literacy literature to put an individual at greater risk and/or vulnerability.

Of concern are females, who were found to be more likely to purchase crypto because of all the hype of social media and who tended not to answer any of the crypto literacy questions correctly. Indigenous peoples were identified to be the most vulnerable to lose their crypto or NFT investment, which is also concerning. Non-Indigenous people also appeared to be attracted to crypto and NFTs because of the attention on social media, while they seemed not to comprehend the importance of holding your own keys (i.e. securing their Bitcoin) and were more likely to be contacted with unsolicited advice. Participants who have English as a second language appeared to have poorer knowledge about the security of their crypto and how to calculate interest.

The findings above highlight the need to educate consumers about how to calculate interest, understand the implications of crypto investments from a tax perspective and how to store their crypto and NFTs safely. We also found that social media was the Number 1 place that participants learned about cryptocurrencies and NFTs, which highlights the need for independent online financial education from trusted sources. We summarise the chi-squared tests results in Table 7.

9 | IMPLICATIONS OF THIS RESEARCH

There is little research on how Australians engage with cryptocurrencies and NFTs, with limited education available to help Australians navigate this area. This research builds conceptual knowledge about who is vulnerable and what vulnerabilities exist for crypto asset and NFT investors. We found that both socioeconomic advantaged or disadvantaged Australians are vulnerable when purchasing cryptocurrencies or NFTs and that social media is a huge influence of individuals who own cryptocurrencies and/or NFTs. Our findings fit with Menzel Baker et al.'s (2005) definition of consumer vulnerability who argue that anyone can be vulnerable, and that vulnerability is not based solely on individual characteristics.

TABLE 7 Summary of participants who were vulnerable when purchasing cryptocurrency or NFTs.

	NFTs purchase	Concerns over the Crypto Storage	Lost funds due to cryptos	Lost funds due to NFTs	Unsolicited advice	Crypto Investment media hype	NFT investment media hype	Insufficient literacy about Q1	Insufficient literacy about Q2	Insufficient literacy about Q3	Insufficient IT literacy about crypto
Gender											
Female						×		×		×	
Male			×						×		
Indigeneity											
Indigenous			×	×							
Non-indigenous	×				×	×		×			
Educational level											
University	×	×									
TAFE		×									
High school or below										×	
Employment condition											
Full-time	×										
Part-time or casual										×	
No										×	×
Living condition											
Living on own			×								
Living with others											

(Continues)

TABLE 7 (Continued)

NFTs purchase	Concerns over the Crypto Storage	Lost funds due to cryptos	Lost funds due to NFTs	Unsolicited advice	Crypto Investment media hype	NFT investment media hype	Insufficient literacy about Q1	Insufficient literacy about Q2	Insufficient literacy about Q3	Insufficient IT literacy about crypto
Housing condition										
Own home with mortgage		×								
Own home no mortgage										
Rent home										
Language spoke at home										
English							×		×	
Other language										

The vulnerabilities include concerns over the security of cryptocurrencies and NFTs, unsolicited advice and limited channels to learn about cryptocurrencies and NFTs, which are all external factors, and insufficient financial and IT literacy in cryptocurrencies and NFTs, which are internal factors. The new knowledge gained from this study may be used to identify what opportunities and/or challenges vulnerable Australians are facing in the online financial world. With the millions of dollars being lost to online crypto investment scams (Australian Government, 2024), targeted online financial education is needed to complement Australians' appetite for crypto assets and NFTs.

With more research needed in this area, we believe the following research questions require further exploration. They include the following:

1. Given the use of social media to promote cryptocurrencies and NFTs, should platform providers and/or individuals who promote these products be required to comply with disclosure obligations in social media posts, outlining in clear and simple language the risks associated with storage, interest rate calculations and tax implications?
2. What is the role of the regulators, ASIC and the ACCC, in protecting consumers from the harmful characteristics of these products, notwithstanding the complexities associated with the international/cross-border and/or decentralised nature of these products?

The above-mentioned research questions are just a couple examples as there are many opportunities to build knowledge about crypto assets and NFTs to help consumers effectively navigate these alternative financial products. Our research identified internal and external factors of vulnerabilities and identified who is vulnerable when purchasing and owning cryptocurrencies and/or NFTs in Australia.

AUTHOR CONTRIBUTIONS

Levon Blue: Conceptualization; investigation; funding acquisition; writing – original draft; writing – review and editing; project administration. **Congcong Xing:** Methodology; software; formal analysis; writing – review and editing. **Thu Pham:** Writing – review and editing; project administration.

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ENDNOTES

¹ In Australia, the Australian Taxation Office treats cryptocurrency purchases as a capital event, meaning that the sale will result in a capital gain or loss of asset, which will need to be reported on a tax return.

² Bitcoin private key is a 256-bit number that can be represented in several ways. Example: E9873D79C6D87DC0F-B6A5778633389F4453213303DA61F20BD67FC233AA33262. (Moreland, 2022, para 21).

³ "At its core, blockchain is a distributed digital ledger that stores data of any kind." (Rodeck, 2022, para 2).

⁴Private keys are important for the security of cryptocurrency. “As per the popular saying in the crypto world, ‘not your keys, not your coins’, having private keys is essential to have complete control over your funds. When you have private keys, you can access, spend, and transfer funds as per your needs, without depending on any third-party or intermediary platform.” (Moreland, 2022, para 12).

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APPENDIX 1

Survey questions

DEVELOPING FINANCIALLY CAPABLE CITIZENS IN AN ERA OF FINANCE APPS AND CRYPTOCURRENCIES

Screening questions:

1. What is your age range?
 - a. Under 18 (Survey ends)
 - b. 18–24
 - c. 25–34
 - d. 35–44
 - e. 45–54
 - f. 55–64
 - g. 65 or above
2. Have you ever bought cryptocurrencies, non-fungible tokens (NFTs), and/or used finance apps such as banking apps, buy now pay later and investing apps? (Please select “yes” for all options if applicable.)
 - a. Yes to cryptocurrencies (go to cryptocurrencies questions and NFTs and finance apps questions will not be displayed. If this option is NOT chosen, the survey will end.)
 - b. Yes to NFTs (go to cryptocurrencies and NFTs questions)
 - c. Yes to finance apps (go to finance apps questions)
 - d. No to all (survey ends)

Note: If both option a, b and c are selected, NFTs, cryptocurrencies and finance apps questions will be displayed.

3. How do you define your living situation?
 - a. Living on my own (skip to Question 7).
 - b. Living with others (then ask Question 4, 5 and 6 below).
 - c. Prefer not to say (then ask Question 4, 5 and 6 below).

Note: Only ask Q4, 5 and 6 if b or c is selected in Q3.
4. Do you share your financial decision making and/or finances with someone else?
 - a. Yes (go to Question 5)
 - b. No (go to cryptocurrencies or finance apps questions depend on which option participant chose in Question 2)
 - c. Prefer not to say (go to cryptocurrencies or finance apps questions depend on which option participant chose in Question 2)

5. Who do you share your financial decision making and/or finances with?
- a. Spouse (go to Question 6)
 - b. Partner (go to Question 6)
 - c. Other (please specify): _____ (go to cryptocurrencies or finance apps questions depend on which option participant chose in Question 1)
 - d. Prefer not to say (go to cryptocurrencies or finance apps questions depend on which option participant chose in Question 1)

6. Who purchased and/or uses the: _____?

	I did	My spouse/partner did	Both of us have	Other	Prefer not to say
NFTs					
Cryptocurrencies					
Finance apps					

- NFTs option only appears if participant chose “Yes to NFTs” in Q2.
- Cryptocurrencies option only appears if participant chose “Yes to cryptocurrencies” in Q2.
- Finance apps option only appears if participant chose “Yes to finance apps” in Q2.

Questions about cryptocurrencies

7. What cryptocurrency(ies) do you currently hold/own? (Please select all that apply.)
- a. Bitcoin
 - b. Ethereum
 - c. Binance Coin
 - d. Cardano
 - e. Solana
 - f. Other (please specify): _____
8. How did you acquire cryptocurrencies? (Please select all that apply.)
- a. Purchased off an exchange
 - b. Received from someone
 - c. Selling items in video games
 - d. Other (please specify): _____

9. What made you want to buy/hold: _____? (Please select all that apply.)

	Potential investment	Long-term store of value	Seeking investment in decentralised currencies	To purchase NFTs	All the hype on social media	Other	Prefer not to say
Cryptocurrencies							
NFTs							

- NFTs option only appears if participant chose “Yes to NFTs” in Q2.

10. Where do you store your: _____? (Please select all that apply.)

	Software wallet (including mobile, desktop and web)	Hardware wallet	On the exchange	Other	Not sure	Prefer not to say
Cryptocurrencies						
NFTs						

- NFTs option only appears if participant chose “Yes to NFTs” in Q2

11. Do you have any concern about the storage of your: _____?

	Yes	No
Cryptocurrencies		
NFTs		

- NFTs option only appears if participants chose “Yes to NFTs” in Q2.
- If yes is selected, then Q12 will be asked:

12. What are your concerns related to?

- Cryptocurrencies: _____
- NFTs: _____

NFTs option only appears if participant chose “Yes to NFTs” in Q2.

13. Have you ever lost your: _____ due to storage issues?

	Yes	No	Prefer not to say
Cryptocurrencies			
NFTs			

- NFTs option only appears if participants chose “Yes to NFTs” in Q2.
- If yes is selected, then Q22 will be asked:

14. How much did you lose?

	Under \$1000	Over \$1000	Prefer not to say
Cryptocurrencies			
NFTs			

15. How would you rate your technology skills?

	Below average	Average	Above average	Prefer not to say	N/A
NFTs					
Purchasing, selling and storing cryptocurrencies					
Finance apps					

- NFTs option only appears if participant chose “Yes to NFTs” in Q2
- Finance apps option only appears if participant chose “Yes to Finance apps” in Q2.

- 16. Have you ever sought financial advice about cryptocurrencies?
 - a. Yes
 - b. No
 - c. Not sure
 - d. Prefer not to say
- 17. Have you ever sought financial advice about investments including superannuation?
 - a. Yes
 - b. No
 - c. Not sure
 - d. Prefer not to say

18. What is the total value of the _____ you own?

		Between \$1000 and \$4999	Between \$5000 and \$9999	Between \$10,000 and \$49,999	Over \$50,000	Prefer not to say
Cryptocurrencies	<\$1000					
NFTs						

- NFTs option only appears if participants chose “Yes to NFTs” in Q2.

- 19. Do you believe the saying “not your keys, not your coins”?
 - a. Yes
 - b. No
 - c. Not sure
 - d. Prefer not to say
- 20. If Bitcoin was worth \$40,000 when you purchased and 12 months later, it was worth \$80,000. How much has Bitcoin increased by?
 - a. 2000%
 - b. 200%
 - c. 50%
 - d. 20%
 - e. Not sure
 - f. Prefer not to say
- 21. Is it true that selling your cryptocurrencies will result in a taxable event (i.e. capital gains or capital loss that needs to be reported to the Australian Taxation Office)?
 - a. True
 - b. False
 - c. Not sure
 - d. Prefer not to say
- 22. Have you ever sold your cryptocurrency?
 - a. Yes
 - (i) Why did you sell your cryptocurrency? (Please select all that apply.)
 - 1. Because I made a profit
 - 2. Because I made a loss
 - 3. Because I needed the money

4. Because I wanted to buy a different cryptocurrency

5. Other (please specify):

b. No

23. Did you pay capital gains taxes on the sale of your cryptocurrencies?

a. Yes

b. No

c. Not sure

d. Other (please specify): _____

e. Prefer not to say

24. Select the statement that best describe your plans with your: _____. (Please select all that apply.)

Long-term investment	Short-term investment	Continuing to acquire more cryptocurrencies	No plans to buy anymore	Planning on selling soon	None of the above are relevant to me	Other	Prefer not to say
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Crypto-currencies

NFTs

- NFTs option only appears if participant chose “Yes to NFTs” in Q2.

25. Who do you discuss: _____ with? (Please select all that apply.)

Friends/colleagues	Family/partner	Online communities	Other	Prefer not to say
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Cryptocurrencies

NFTs

- NFTs option only appears if participant chose “Yes to NFTs” in Q2.

26. Have you ever been approached by someone offering unsolicited advice and/or investments relating to cryptocurrencies?

a. Yes

(i) How did they contact you? (Please select all that apply.)

1. E-mail

2. Text

3. Phone

4. Direct messages (DM)

5. Other (please specify):

b. No

Questions about finance apps

27. What finance apps do you use? (Please select all that apply) (this question will NOT be displayed if Q6—the option “My spouse/partner did” is selected)

a. Banking apps

b. Investing

c. Budgeting

- d. Buy now, pay later
- e. Wages advance
- f. Other (please specify):

28. Select the statement that best describe how finance apps impacted your financial situation. (Please select all that apply.)
- a. Finance apps give me greater control over my financial future
 - b. Provide me with the ability to absorb a financial shock
 - c. Give me greater financial freedom to make choices to enjoy life
 - d. Give me the ability to meet my financial goals
 - e. None of the above are relevant to me
 - f. Prefer not to say

29. How did you learn about: _____? (Please select all that apply.)

Social media	School/TAFE/ university	Family or friends	Financial planner/ advisor	Other	Prefer not to say
NFTs					
Cryptocurrencies					
Finance apps					

- If social media option is selected, then Q29 will be asked:

30. Which social media platforms did you learn about:_____? (Please select all that apply.)

Tiktok	YouTube	Twitter	Instagram	Facebook	LinkedIn	Other
NFTs						
Cryptocurrencies						
Finance apps						

- NFTs option only appears if participant chose “Yes to NFTs” in Q2.
- Cryptocurrencies option only appears if participant chose “Yes to cryptocurrencies” in Q2.
- Finance apps option only appears if participants chose “Yes to finance apps” in Q2.

Questions about NFTs, cryptocurrencies and finance apps

31.Where would you recommend someone interested in learning more about: _____ go?

- NFTs: _____
- Cryptocurrencies: _____
- Finance apps: _____

32. How do you prefer to learn about NFTs, cryptocurrencies and/or finance apps? (Please select all that apply)?

- a. Online videos/courses
- b. Social media platforms on:
 - 1. Tiktok
 - 2. YouTube
 - 3. Twitter
 - 4. Instagram

- 5. Facebook
- 6. LinkedIn
- 7. Other (please specify): _____
- c. School/ TAFE/ University
- d. Financial planner/advisors
- e. Other (please specify):

Demographic questions

33. What is your gender?
- a. Male
 - b. Female
 - c. Non-binary
 - d. Other
 - e. Prefer not to say
34. Do you identify as:
- a. Aboriginal
 - b. Torres Strait Islander
 - c. Aboriginal and Torres Strait Islander
 - d. Neither Aboriginal or Torres Strait Islander
 - e. Prefer not to say
35. Are you currently employed?
- a. Yes—Full-time (go to Q35)
 - b. Yes—Part-time and/or casually (go to Q35)
 - c. No (go to Q34)
 - d. Prefer not to say (go to Q34)
36. Are you currently studying?
- a. Yes—Full-time
 - b. Yes—Part-time and/or casually
 - c. No
 - d. Prefer not to say
37. What is your highest level of education?
- a. Some high school
 - b. High school
 - c. TAFE
 - d. University
 - e. Other (please specify)
38. What is the postal code where you live?
- _____
39. Do you currently _____?

- a. Own your home with a mortgage
 - b. Own your home without a mortgage
 - c. Rent your home
 - d. Other (please specify): _____
 - e. Prefer not to say
40. Do you speak another language at home other than English?
- a. Yes (please specify the language):
 - b. No
 - c. Prefer not to say

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Associate Professor **Levon Blue** is the Indigenous Research Development Coordinator at The University of Queensland in the Office of the Deputy-Vice-Chancellor Indigenous Engagement. Levon was awarded a PhD in 2016, which focussed on financial literacy education practices in a First Nation community in Canada. She is a member of Beausoleil First Nation in Canada and lives in Queensland, Australia. Levon's research area includes financial literacy education and higher education with Indigenous peoples.

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Dr **Thu Pham** is originally from Vietnam and currently lives and works on Turrbal and Yugara Country in Queensland. She is a senior research assistant at Griffith University. Her research areas include leadership in higher education and Indigenous students' success. She completed her doctoral degree in Education at the Queensland University of Technology, Australia, in 2016. Her doctoral research study focussed on leadership to support quality improvement in Vietnamese higher education. She is currently working on several research projects focussing on Aboriginal and Torres Strait Islander people in Australia regarding the impact of digital inclusion challenges for low-income Indigenous families living on Mornington Island, a remote region in Queensland and the social determinants of higher education impact Indigenous students' attainment. Recently, she has published a Springer Brief titled "Higher Degree by Research: Factors for Indigenous student success."